

Chapter 5: outline

5.1 introduction

5.2 routing protocols

- link state
- distance vector

5.3 intra-AS routing in the Internet:

- RIP
- OSPF

5.4 inter-AS routing in the Internet: BGP

Routing protocols

Routing protocol goal: determine “good” paths (equivalently, routes), from sending host to receiving host, through network of routers

- path: sequence of routers packets will traverse in going from given initial source host to given final destination host
- “good”: “least cost”, “fastest”, “least congested”
- routing: a “top-10” networking challenge!

Components of a Routing Algorithm

- a procedure for **sending and receiving** reachability information about a network to other routers
- a procedure for **calculating optimal** routes
 - Routes are calculated using a shortest path algorithm (least “**cost**”)
- a procedure for **reacting to and advertising** topology changes

Routing algorithm classification

Q: global or local information?

global:

- all routers have complete topology, link cost info
- “link state” algorithms

local:

- router knows physically-connected neighbors, link costs to neighbors
- iterative process of computation, exchange of info with neighbors
- “distance vector” algorithms

Q: quasi-static or dynamic?

quasi-static:

- routes change slowly over time

dynamic:

- routes change more quickly
 - periodic update
 - in response to link cost changes

Two Shortest Path IGP Routing Algorithms

Distance Vector Routing

- each node knows the distance (cost) to its directly connected neighbors
- a node periodically sends a list of routing updates to its neighbors
- if all nodes update their distances to destinations using neighbor information, the routing tables eventually converge

Link State Routing

- each node knows the distance (cost) to its directly connected neighbors
- the distance information is flooded to all nodes in the network
- each node calculates the routing tables independently using a network map (topology) created by the node using the global information it received

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Routing Information Protocol (RIP)

- “open”: publicly available
- uses distance vector algorithm
 - routing table dissemination
 - only link cost to neighbors at each node
 - route computation using Bellman Ford’s algorithm
- router advertises its routing table to all its neighbors
 - carried in RIP messages over UDP
 - routing table entries give cost from that node (source) to all other nodes (destinations) in AS

RIP: Basic principles

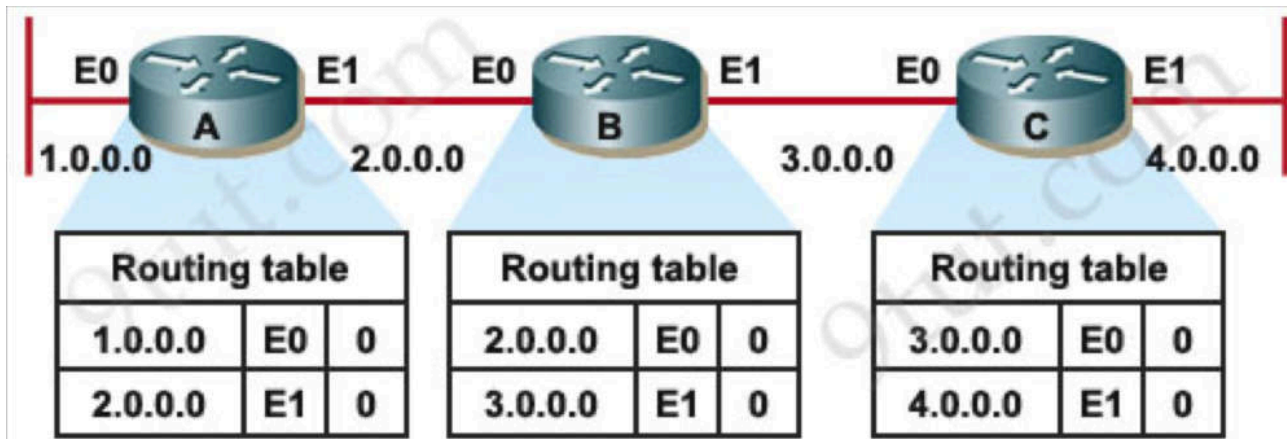
- periodically, every router sends to its neighbors a complete list of its routes to all destinations within an AS
- list contains pairs of: **destination**, **distance**
- receiver replaces/updates entries in its routing table if routing through a **neighbor** costs less than the current route in its table

Rip Example

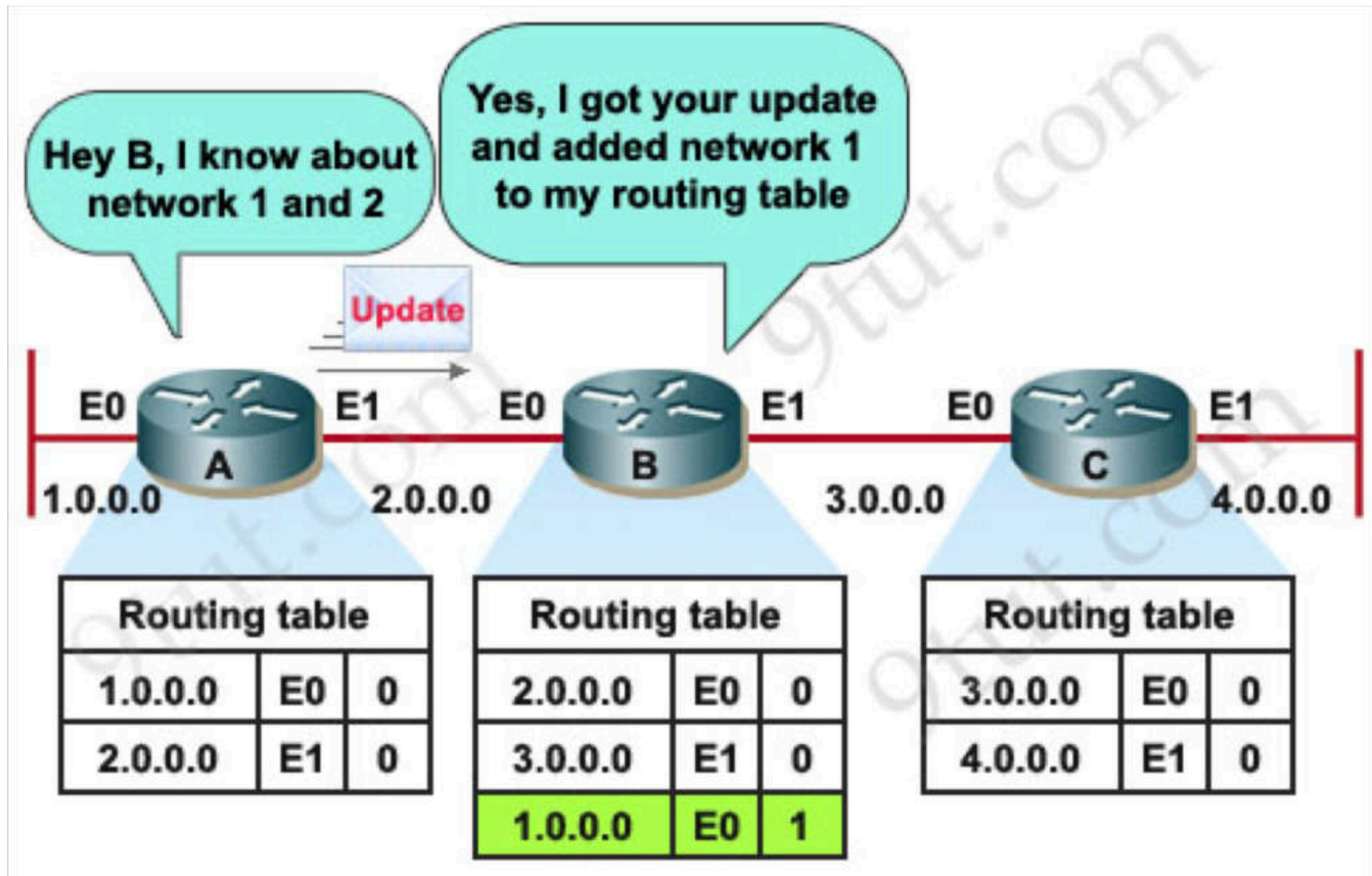
assume:

- link cost is 1 on all hops
- all updates occur simultaneously

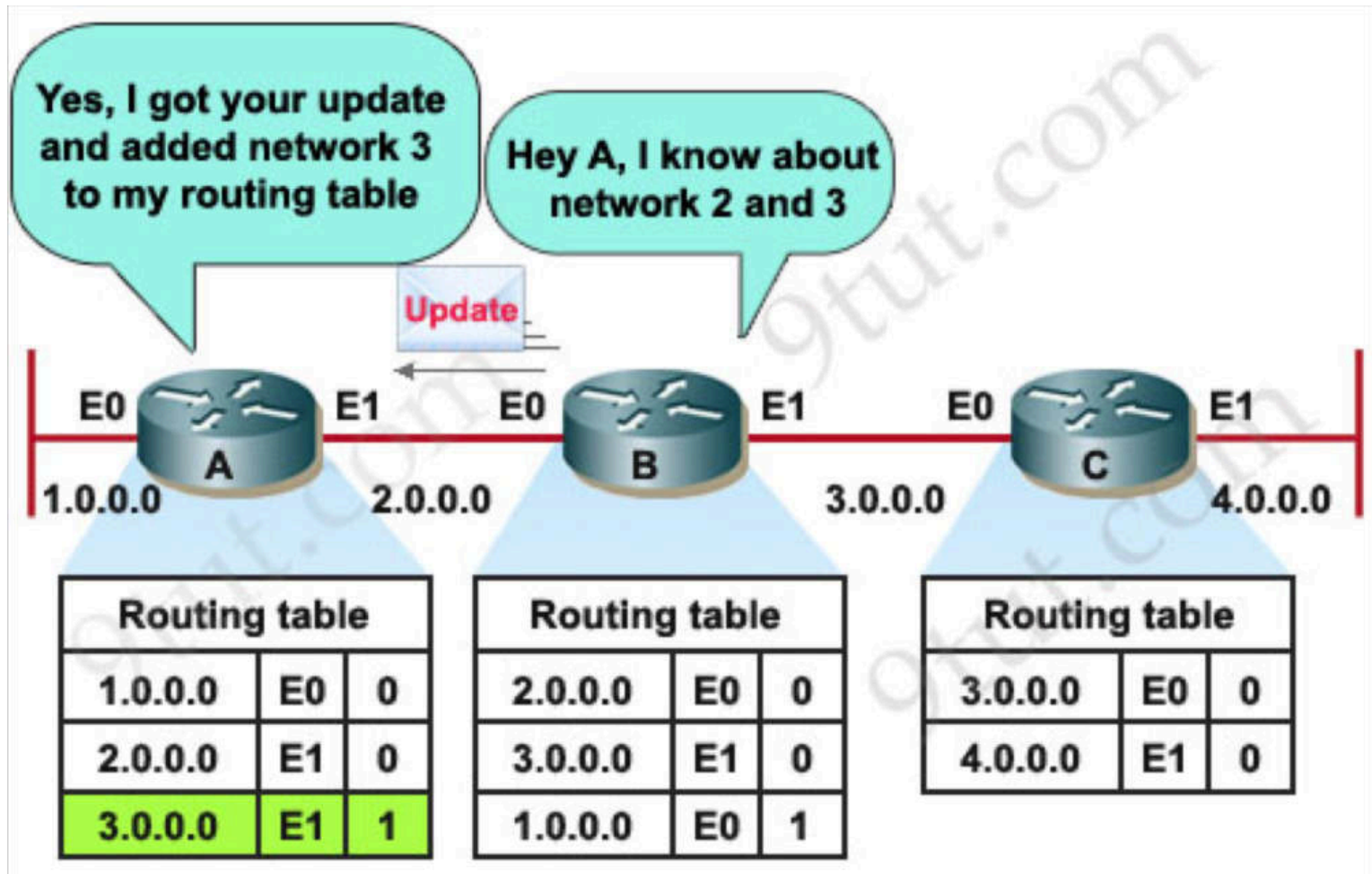
initially each router only knows its directly connected interfaces -> cost = 0



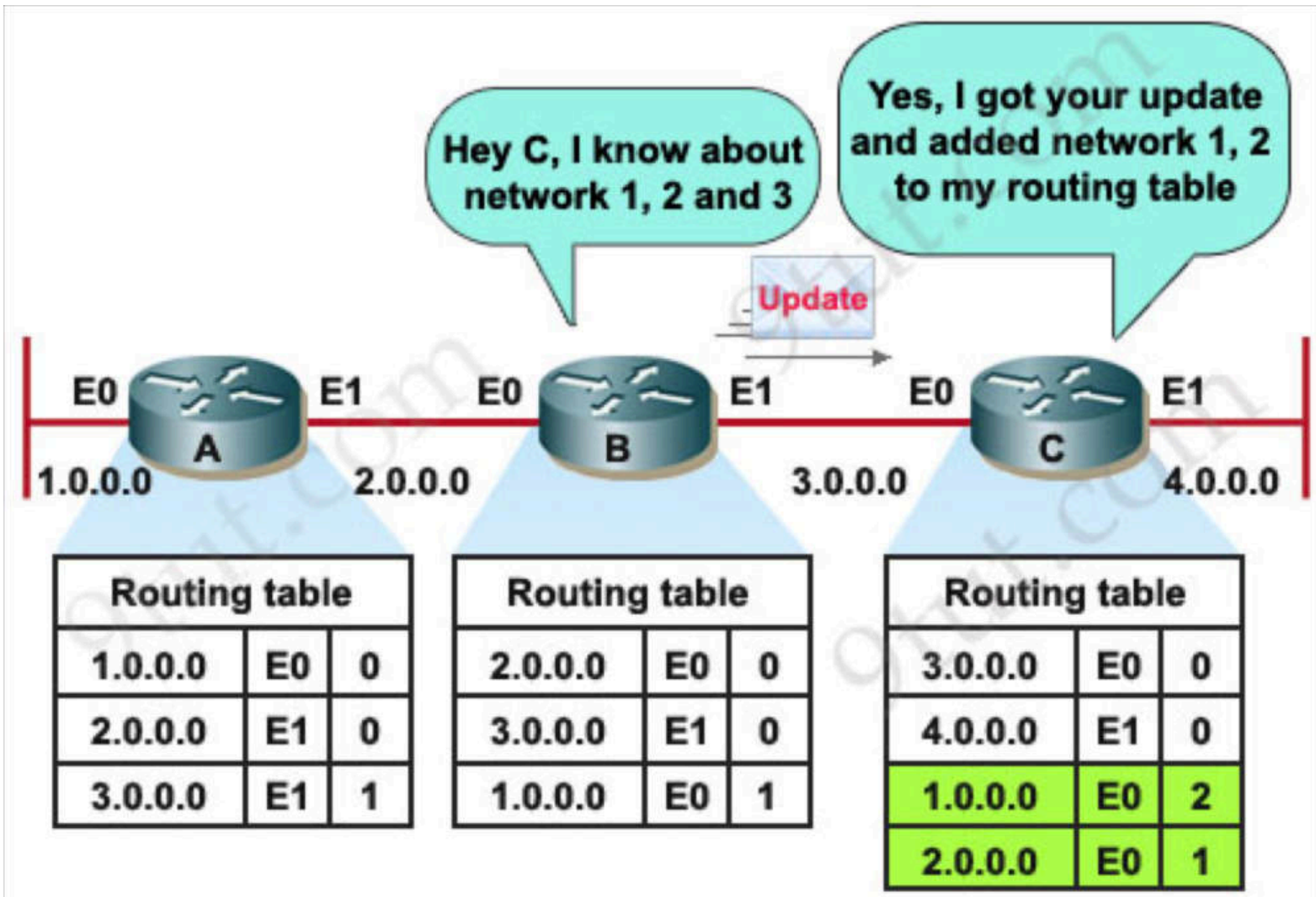
After First Update



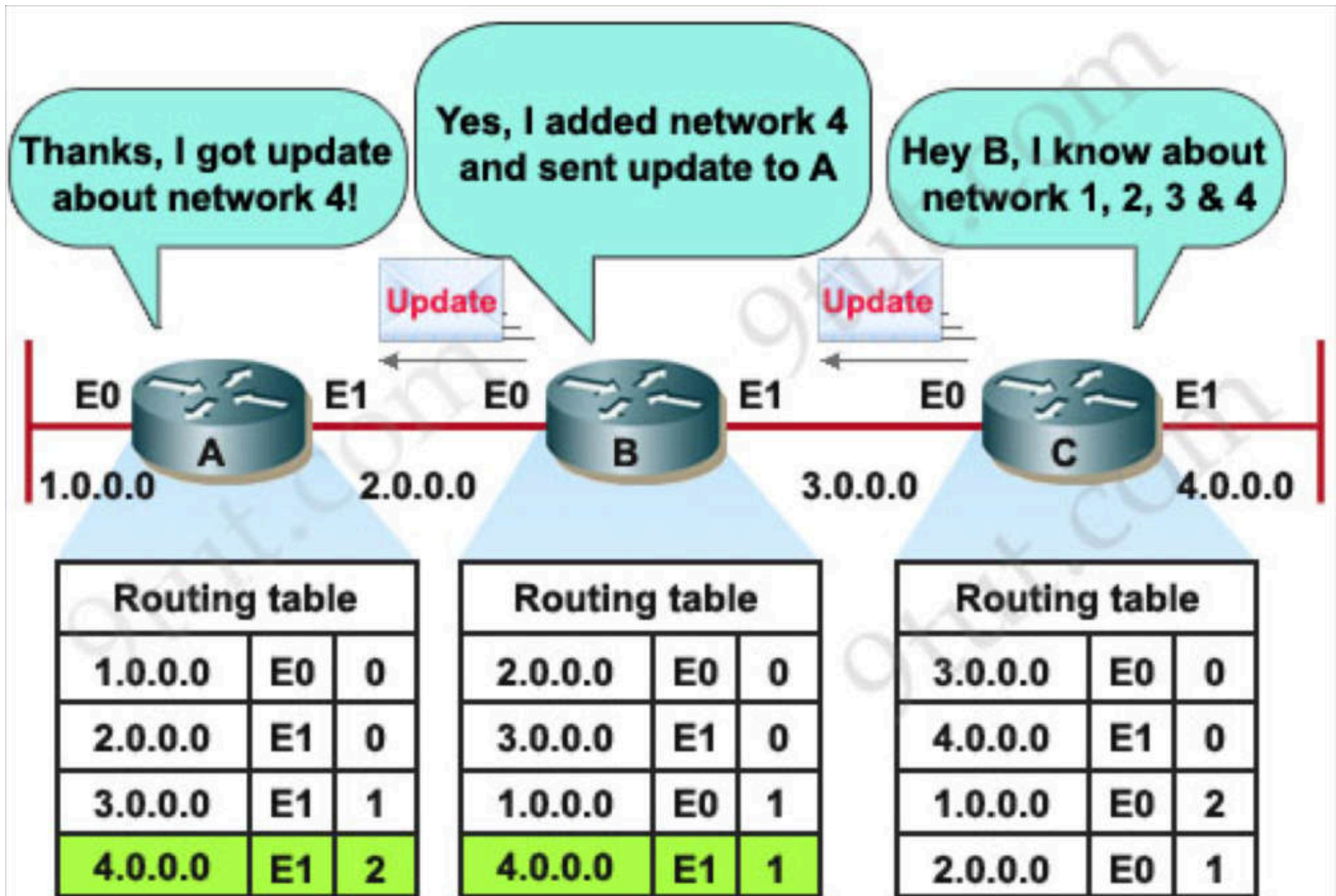
After Second Update



After Third Update



Last Update for Convergence



Convergence and Loops

- Distance Vector Protocols are subject to **loop formations** because of the **myopic view** of each router.
 - routers only hear from neighbors and use that to create a global connectivity map.
- when changes occur, they are broadcast but take a while to propagate and during that time **cycles** can form.
- one particular problem is the **count to infinity** problem, where updates bounce back and forth and the distance or cost creeps up in value.
- to counter that, a **maximum value** is set that once it is reached, the destination is considered to be **unreachable** and the route is removed from the routing table.

RIP Summary and demise

- low overhead – fully distributed ... BUT
- slow convergence
- low overhead
- limited to 15 hops (max path cost $\rightarrow$ infinity = 16)
- only uses local information from immediate neighbors for routing decisions - relies on propagation of information for global view of network – cycle formations
- No longer used – Rest in Peace (RIP)

OSPF (Open Shortest Path First)

- “open”: publicly available
- uses link-state algorithm
 - link state packet dissemination
 - topology map at each node
 - route computation using Dijkstra’s algorithm
- router floods OSPF link-state advertisements to all other routers in *entire* AS
 - carried in OSPF messages directly over IP (rather than TCP or UDP – protocol type 89)
 - link state: for each attached link

OSPF: Basic principles

- routers establish a relationship (“*adjacency*”) with neighbors
- each router generates *link state advertisements (LSAs)* which are distributed to all “*adjacent*” routers (after routers have established adjacencies).

LSA = (link id, state of the link, cost, neighbors of the link)

- each router maintains a database (LSDB). consists of all received LSAs (*topological database* or *link state database*), which describes the network as a graph with weighted edges
- each router uses its link state database to run a shortest path algorithm (Dijkstra’s algorithm) to produce the shortest path to each network

Operation of a Link State Routing protocol

